

02_zerocounts_overlap

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This analysis identifies sgRNAs with zero or low counts across samples, and evaluates their overlap between conditions. This informs filtering decisions for downstream analysis and lets us see how randomly dispersed sgRNA dropouts are.

Load data and identify sgRNAs with zero counts

```
counts <- read.delim("data_raw/allnonorm.count.txt")
plasmid <- read.delim("data_raw/R1formattedH1.count.txt")
zero_T0 <- counts %>% filter(T0 == 0) %>% pull(sgRNA)
zero_D7 <- counts %>% filter(D7 == 0) %>% pull(sgRNA)
zero_LL <- counts %>% filter(LL == 0) %>% pull(sgRNA)
zero_UL <- counts %>% filter(UL == 0) %>% pull(sgRNA)
zero_plasmid <- plasmid %>% filter(L1 == 0) %>% pull(sgRNA)
```

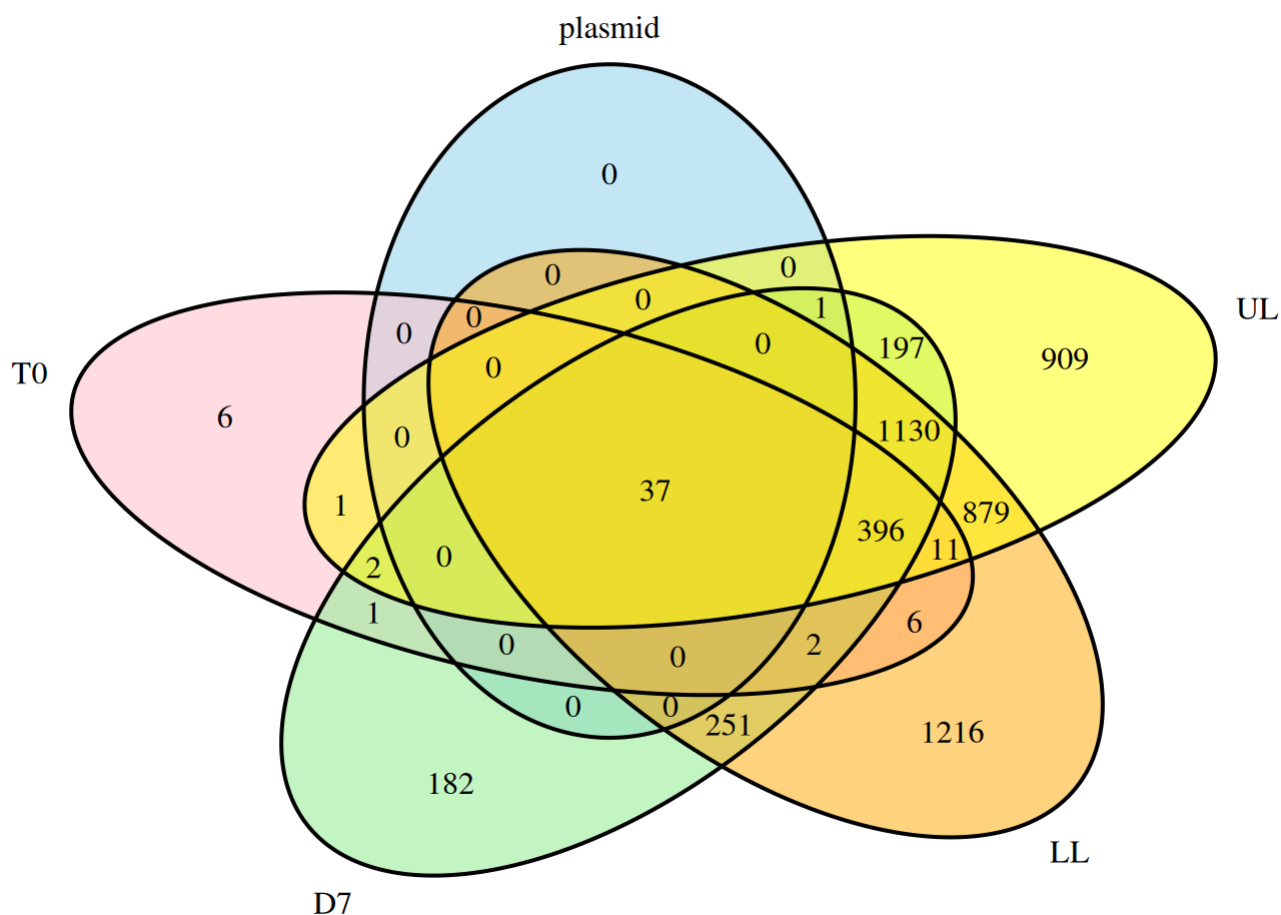
Including Plots

Visualise overlaps as venn diagrams

```
venn.plot <- venn.diagram(
  x = list(
    plasmid = zero_plasmid,
    T0 = zero_T0,
    D7 = zero_D7,
    LL = zero_LL,
    UL = zero_UL ),
  filename = NULL,
  fill = c("skyblue", "pink", "lightgreen","orange", "yellow"),
  alpha = 0.5,
  main = "Zero-count sgRNAs overlap")

grid::grid.draw(venn.plot)
```

Zero-count sgRNAs overlap

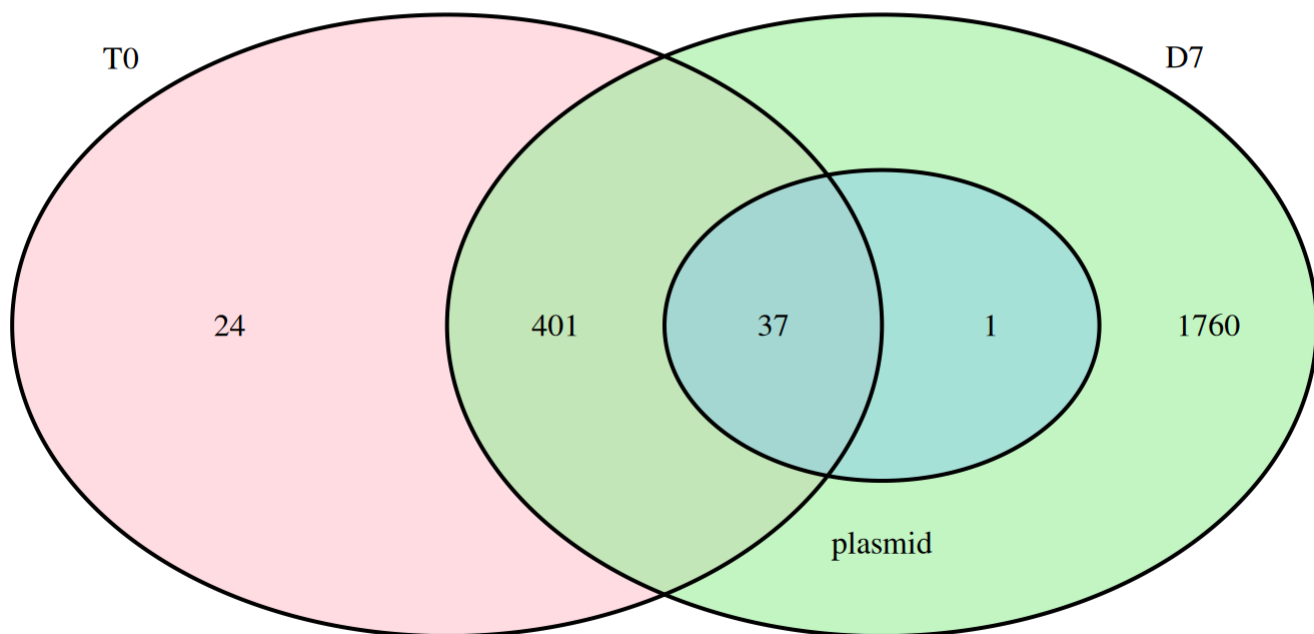


Now look at controls only

```
venn.plot <- venn.diagram(
  x = list(
    plasmid = zero_plasmid,
    T0 = zero_T0,
    D7 = zero_D7 ),
  filename = NULL,
  fill = c("skyblue", "pink", "lightgreen"),
  alpha = 0.5,
  main = "Zero-count sgRNAs overlap (controls)")

grid::grid.draw(venn.plot)
```

Zero-count sgRNAs overlap (controls)



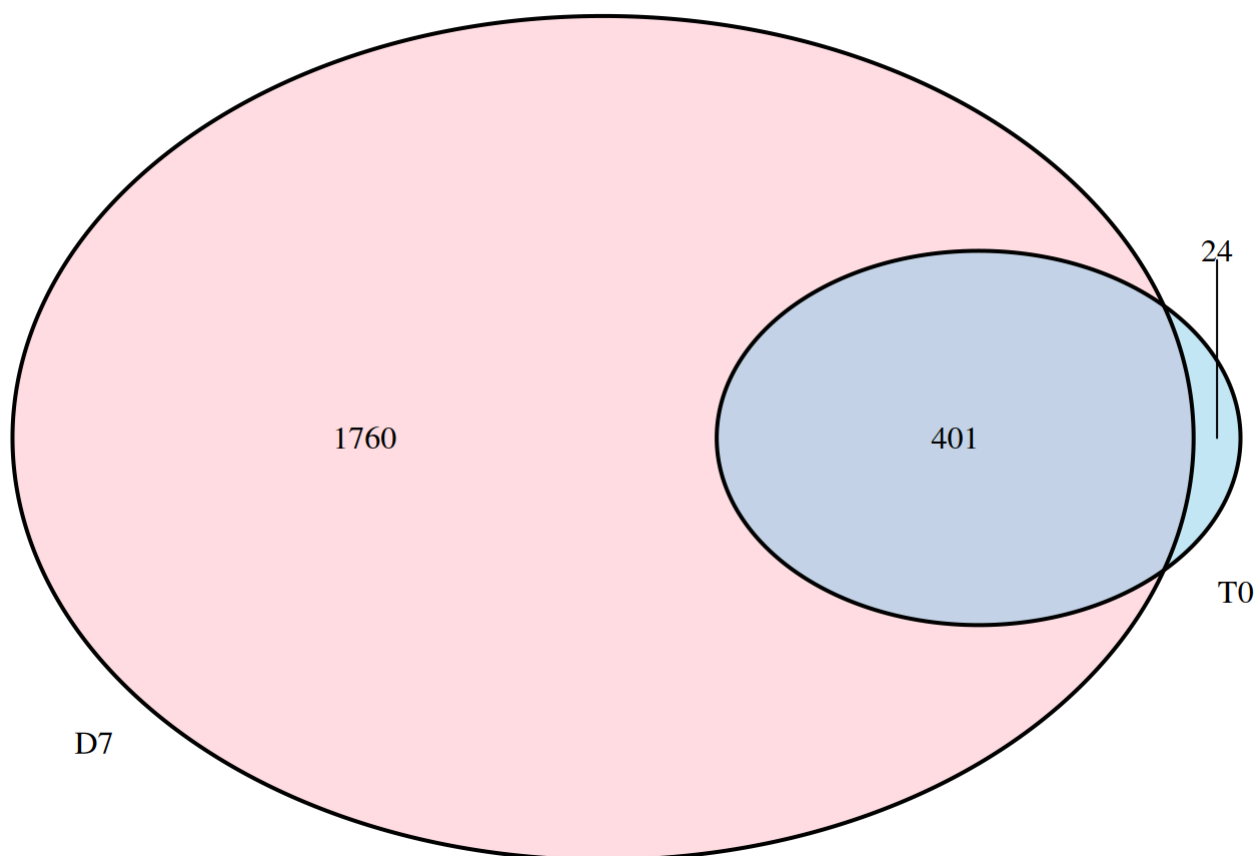
```
#investigate discrepancy
setdiff(zero_plasmid, zero_T0)
```

```
## [1] "MASP2_-_11107265.23-P1P2"
```

Remove plasmid zero-count guides and overlap samples again after filtering:

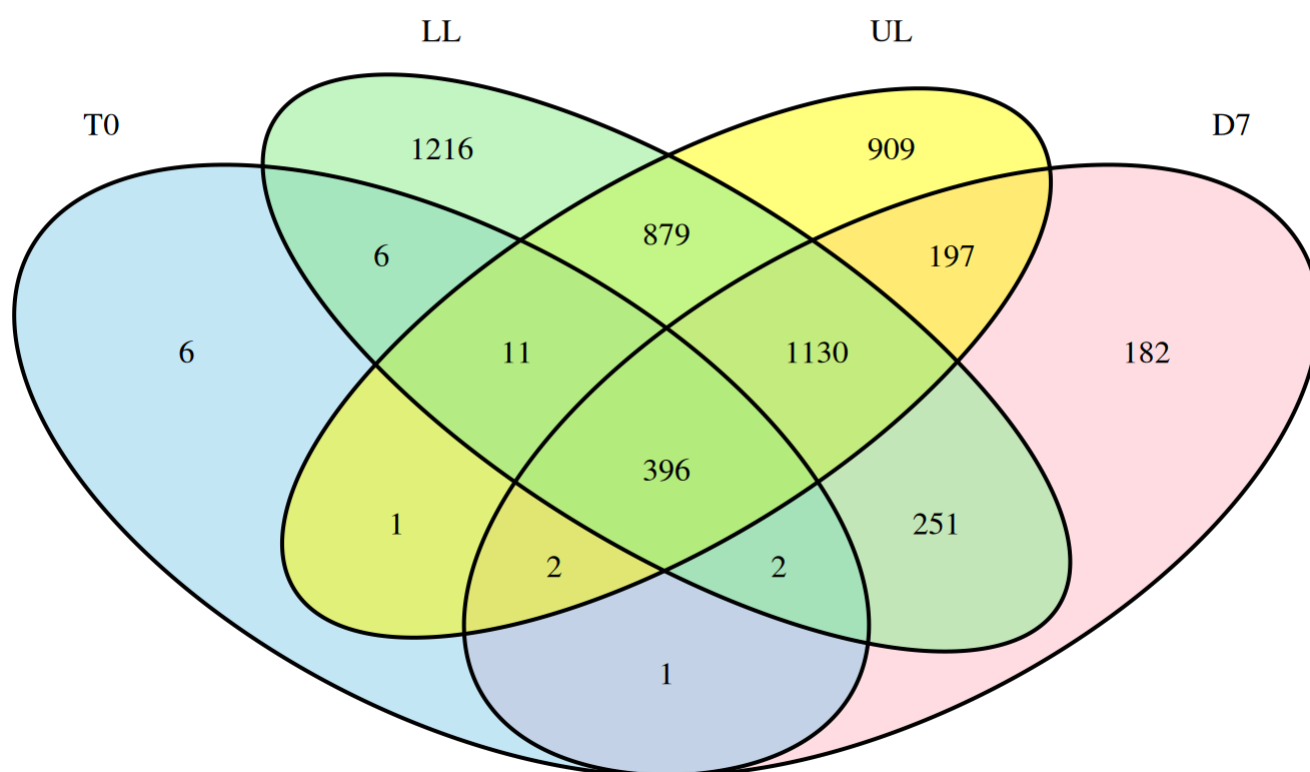
```
T0 <- setdiff(zero_T0, zero_plasmid)
D7 <- setdiff(zero_D7, zero_plasmid)
LL <- setdiff(zero_LL, zero_plasmid)
UL <- setdiff(zero_UL, zero_plasmid)
#now overlap
venn.plot <- venn.diagram(
  x = list(T0 = T0, D7 = D7),
  filename = NULL,
  fill = c("skyblue", "pink"),
  alpha = 0.5,
  main = "Zero-count overlap (filtered controls)")
grid::grid.draw(venn.plot)
```

Zero-count overlap (filtered controls)



```
venn.plot <- venn.diagram(  
  x = list(T0 = T0, D7 = D7, LL = LL, UL = UL),  
  filename = NULL,  
  fill = c("skyblue", "pink", "lightgreen", "yellow"),  
  alpha = 0.5,  
  main = "Zero-count overlap (filtered)")  
grid::grid.draw(venn.plot)
```

Zero-count overlap (filtered)



identify guides missing across all samples (these will be uninformative). Annotate these missing guides

```
#identify sgRNAs missing across all samples
T0 <- setdiff(zero_T0, zero_plasmid)
D7 <- setdiff(zero_D7, zero_plasmid)
LL <- setdiff(zero_LL, zero_plasmid)
UL <- setdiff(zero_UL, zero_plasmid)
zero_in_all <- Reduce(intersect, list(T0, D7, LL, UL))
length(zero_in_all)
```

```
## [1] 396
```

```
#annotate
missing_guides_info <- plasmid[ plasmid$sgRNA %in% zero_in_all, c("sgRNA", "Gene") ]
missing_per_gene <- aggregate(sgRNA ~ Gene, data = missing_guides_info, FUN = length)
head(missing_per_gene)
```

```
##      Gene sgRNA
## 1 AADACL2    1
## 2  AARS     5
## 3  AARS2     1
## 4  ABCE1     2
## 5  ACACA     3
## 6  ACADS     1
```

Repeat analysis using <10 reads threshold and identify guides missing across all samples. This <10 reads threshold was used to define effectively absent sgRNAs, as low counts are more susceptible to stochastic noise and can be misleading

```
zero_T0 <- counts %>% filter(T0 < 10) %>% pull(sgRNA)
zero_D7 <- counts %>% filter(D7 < 10) %>% pull(sgRNA)
zero_LL <- counts %>% filter(LL < 10) %>% pull(sgRNA)
zero_UL <- counts %>% filter(UL < 10) %>% pull(sgRNA)
zero_plasmid <- plasmid %>% filter(L1 < 10) %>% pull(sgRNA)
```